

Name of the student:

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2023A3 PS10386

Question	Marks	Score
1	7	07
2	8	08
Total:	15	15

Question 1

Logical function $(A+B)(C+D)$ is to be implement using minimum number of two input NOR gates. Assume that load capacitance is 60 units and input capacitance (on every input) is 25 units. Assume the PMOS resistance is 2 times that of NMOS for same size. If $R=12\text{ k}\Omega$ and $C=0.1\text{ fF}$ for given technology, calculate minimum delay from input to output and size of each transistor is each stage.

Normalized delay	9.164	2
Actual delay	32.99 ps	1
Stage 1 NMOS size	5	1
Stage 1 PMOS size	20	1
Stage 2 NMOS size	7.75	1
Stage 2 PMOS size	31	1

Question 2

Logical function $(A+B)(C+D)$ is to be implement using complex OAI22 gate followed by inverter. Assume that load capacitance is 60 units and input capacitance (on every input) is 25 units. Assume the PMOS resistance is 2 times that of NMOS for same size. If $R=12\text{ k}\Omega$ and $C=0.1\text{ fF}$ for given technology, calculate minimum delay from input to output and size of each transistor is each stage.

Normalized delay	9.382	3
Actual delay	33.78 ps	1
Stage 1 NMOS size	8.33	1
Stage 1 PMOS size	16.66	1
Stage 2 NMOS size	9.13	1
Stage 2 PMOS size	18.26	1